Reg. No. | | | | | | | | | | |

[22CSE133]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER - 2022

INTRODUCTION TO PYTHON PROGRAMMING FOR AI-DE

Time: 3 Hours **Maximum Marks: 100**

Instructions:

PART-A (5×2=10)

ANSWER ALL QUESTIONS

1.	**a.**	Define a python expression with example.	L-1	CO1	**02**
	b.	Accept an input from the user and compose an if statement to print the value if the entered value is less than 25.	L-2	CO2	**02**
	c.	Select and enter values to a list of integer data type with a length of 5, add a value at the end to the existing list. Print the modified list.	L-2	CO3	**02**
	d.	Write a program to print "hello" 5 times using for loop.	L-3	CO4	**02**
	e.	Interpret a 2×3 matrix and print.	L-2	CO5	**02**
		PART-B (6×5=30 Marks)			
2.	**a.**	Differentiate between list and tuple with examples	L-3	CO1	**05**
		OR			
	b.	Differentiate between for loop and while loop	L-3	CO2	**05**
3.	**a.**	Write a program to compute the factorial of a given number.	L-2	CO2	**05**
		OR			
	b.	Solve a program that print the following Triangle Number pattern, using nested for loop. 6 5 4 3 2 1 5 4 3 2 1 4 3 2 1 3 2 1 2 1 1	L-3	CO2	**05**
4.	**a.**	Write a program that takes date and time inputs from the keyboard. This input is then used to display n dates in a sequential order with the same time as taken in the inputs. Format of the data input 2022-12-3 9:00	L-3	CO3	**05**
		OR			
	b.	Write a program that takes nput for 2*3 matrix ad multiply all elements of te matrix by 10 and print the same.	L-3	CO3	**05**
5.	**a.**	Write a program that takes in random inputs for a 2×3 matrix. Use the math operation multiplication of 10. program has to print the final results.	L-3	CO4	**05**

		OR			
	b.	Explain Scalar Values, Vectors, Matrics with examples.	L-2	CO5	**05**
6.	**a.**	Illustrate that if you have data contained in a Python dict, can you create a Series from it by passing the dict. If yes explain, if no write a program.	L-3	CO3	**05**
		OR			
	b.	Demonstrate if a column in a DataFrame can be retrieved as a Series. If yes explain with anexample, if no explain with an example.	L-3	CO3	**05**
7.	**a.**	Write a program to create a dictionary of character frequencies where the word is entered through the keyboard.	L-3	CO6	**05**
		OR			
	b.	Demonstrate with a program to generate 100 random numbers from 1 to 50, then change first two 10s to -1.	L-3	CO6	**05**
		PART - C (6×10=60 Marks)			
8.	**a.**	Explain Equality and relational operators with 2 examples each.	L-2	CO1	**10**
		OR			
	b.	Explain break and continue with eamples	L-2	CO1	**10**
9.	**a.**	Demonstrate the binary recursive search algorithm with the help of a program giving the steps.	L-3	CO2	**10**
		OR			
	b.	Explain the use of nested loops to print the prime numbers between 2 to 50.	L-2	CO2	**10**
10.	**a.**	Demonstrate a program to write menu driven program to do various list operations such as **i.** append an element, **ii.** Insert an element at a position **iii.** Append a list to the given list .	L-3	CO3	**10**
		OR			
	b.	Demonstrate a program to calculate average marks of n students using a function where the marks are entered by the user.	L-3	CO3	**10**
11.	**a.**	Demonstrate with a program to multiply two matrices using nested loops.	L-3	CO4	**10**
		OR			
	b.	Demonstrate with a program to create numpy array object of 2*3 and print the type of object, dimensions, shape, total number of elements,	L-3	CO4	**10**

		type of elements in the array.			
12.	**a.**	Explain pandas series and dataframe with example.	L-2	CO5	**10**
		OR			
	b.	Demonstrate with a program to create a pandas data frame with column names as state,year and population. **i.** What will happen if you try creating a column that does not exist? **ii.** Suppose a column is empty in Pandas data_frame, how can you modify by assignment. The empty column with the same scalar values in that column.	L-3	CO5	**10**
13.	**a.**	Demonstrate with a program to take two words through the keyboard, create a dictionary of character frequencies print if these words contain the same character with same frequencies.	L-3	CO6	**10**
		OR			
	b.	Demonstrate with a program to take 5 numbers from user, print out two smallest use eval.	L-3	CO6	**10**